

Def. Let S be a regular surface and $T_p S$ be the tangent plane at $p \in S$.
A quadratic form

$$I_p: T_p S \rightarrow \mathbb{R}: w \mapsto \langle w, w \rangle_p \geq 0$$

is called the first fundamental form of S at p .

Computation of the fundamental form.

Given $p \in S$, let $X: U \rightarrow S$ be a parametrization of S containing p .

Let $w \in T_p S$ be given. Take a curve

$$\alpha: (-\varepsilon, \varepsilon) \rightarrow X[U]: t \mapsto X(u(t), v(t)) \quad \text{such that } \alpha(0) = p \text{ and } \alpha'(0) = w.$$

Now,

$$\begin{aligned} I_p(w) &= I_p(\alpha'(0)) = \langle \alpha'(0), \alpha'(0) \rangle_p \\ &= \langle X_u u' + X_v v', X_u u' + X_v v' \rangle \\ &= \langle X_u, X_u \rangle \cdot (u')^2 + 2 \cdot \langle X_u, X_v \rangle \cdot u'v' + \langle X_v, X_v \rangle \cdot (v')^2. \quad (t=0). \end{aligned}$$

In this sense, define

$$E(u_0, v_0) = \langle X_u, X_u \rangle_p, \quad F(u_0, v_0) = \langle X_u, X_v \rangle_p, \quad G(u_0, v_0) = \langle X_v, X_v \rangle_p.$$

Def. A length of parametrized curve $\alpha: I \rightarrow S$ is

$$S(t) = \int_0^t |\alpha'(r)| dr = \int_0^t \sqrt{I(\alpha'(r))} dr = \int_0^t \sqrt{E \cdot (u')^2 + 2F \cdot u'v' + G \cdot (v')^2} dr$$

Def. Let $R \subset S$ be a bounded set contained a image of $X: U \rightarrow S$,

A area of R is:

$$A(R) = \iint_{X^{-1}[R]} |X_u \times X_v| du dv.$$

Examples,

Let $S = \{(x, y, z) \mid x^2 + y^2 = 1\}$ be the cylinder.

Consider a parametrization of S ,

$$X: U \rightarrow \mathbb{R}^3: (u, v) \mapsto (\cos u, \sin u, v)$$

where $U = \{(u, v) \in \mathbb{R}^2 \mid 0 < u < 2\pi, -\infty < v < \infty\}$,

Simply, $X_u = (-\sin u, \cos u, 0)$ and $X_v = (0, 0, 1)$.

So, $E = 1$, $F = 0$, $G = 1$.

Helicoid.

Let $S = \{(r \cdot \cos \theta, r \cdot \sin \theta, a$

Exercises. Sec 2.5.

1. Evaluate the first-fundamental form of the following:

a). $X(u,v) = (a \cdot \sin u \cdot \cos v, b \cdot \sin u \cdot \sin v, c \cdot \cos u)$

b). $X(u,v) = (a \cdot u \cos u, b \cdot u \sin u, u^2)$

Proof.

a). $X_u = (a \cos u \cdot \cos v, b \cos u \cdot \sin v, -c \cdot \sin u)$

$$X_v = (-a \sin u \cdot \sin v, b \sin u \cdot \cos v, 0)$$

$$\begin{aligned} \text{So, } E = \langle X_u, X_u \rangle &= a^2 \cos^2 u \cdot \cos^2 v + b^2 \cos^2 u \cdot \sin^2 v + c^2 \cdot \sin^2 u \\ &= (a^2 \cos^2 v + b^2 \sin^2 v) \cos^2 u + c^2 \cdot \sin^2 u \end{aligned}$$

$$\begin{aligned} F = \langle X_u, X_v \rangle &= -a^2 \cos u \cdot \sin u \cdot \cos v \cdot \sin v + b^2 \cos u \cdot \sin u \cdot \cos v \cdot \sin v \\ &= (-a^2 + b^2) \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v \end{aligned}$$

$$G = \langle X_v, X_v \rangle = a^2 \sin^2 u \cdot \sin^2 v + b^2 \sin^2 u \cdot \cos^2 v$$

Fundamental form of the Monge patch.

Given a differentiable map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, consider the Monge patch

$$X: U \rightarrow \mathbb{R}^3: (u, v) \mapsto (u, v, f(u, v)).$$

Then, the partial derivative is

$$X_u = (1, 0, f_u) \text{ and } X_v = (0, 1, f_v)$$

so, the first-fundamental form is

$$E = \langle X_u, X_u \rangle = 1 + (f_u)^2, \quad F = \langle X_u, X_v \rangle = f_u \cdot f_v, \quad G = \langle X_v, X_v \rangle = 1 + (f_v)^2.$$

Moreover, $\det \begin{pmatrix} E & F \\ F & G \end{pmatrix} = 1 + (f_u)^2 + (f_v)^2 \geq 1 > 0,$

Notation

Denote the Coordinate Patch as

$$X: \mathcal{U} \rightarrow \mathbb{R}^3: (u^1, u^2) \mapsto (f^1, f^2, f^3)$$

Second fundamental form.

Def. Let $X: \mathcal{U} \rightarrow \mathbb{R}^3$ be a coordinate patch.

The second fundamental form of X is:

$$L_{i\bar{j}} \stackrel{\text{def}}{=} \langle X_{i\bar{j}}, n \rangle \quad (1 \leq i, \bar{j} \leq 2)$$

And the Christoffel symbols is

$$\Gamma_{i\bar{j}}^k \stackrel{\text{def}}{=} \sum_{l=1}^2 \langle X_{i\bar{j}}, X_l \rangle \cdot g^{lk}.$$

Where $X_{i\bar{j}} := \frac{\partial^2 X}{\partial u^i \partial v^j}$ and g^{lk} is (k, l) entry of $\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}^{-1}$.

$$n = \frac{X_1 \times X_2}{\|X_1 \times X_2\|} = \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}.$$

Normal and geodesic Curvature.

Def Let $X: U \rightarrow \mathbb{R}^3$ be a patch and

$$\gamma: I \rightarrow \mathbb{R}^3: s \mapsto X(\gamma^1(s), \gamma^2(s))$$

be a unit speed curve whose image is contained in the patch.

Define the Normal Curvature

$$K_n(s) = \langle \gamma''(s), n(s) \rangle$$

and the Geodesic Curvature

$$K_g(s) = \langle \gamma''(s), n(s) \times T(s) \rangle$$

So that $K(s) \cdot N(s) = T'(s) = \gamma''(s) = K_n(s) \cdot n(s) + K_g(s) \cdot n(s) \times T(s)$.

$$\text{And } K^2 = K_n^2 + K_g^2$$

Normal of Monge patch.

Let $X: U \rightarrow \mathbb{R}^3: (u, v) \mapsto (u, v, f(u, v))$

$$\begin{pmatrix} 1 & 0 & f_u \\ 0 & 1 & f_v \end{pmatrix}$$

Then,

$$n = \frac{X_u \times X_v}{\|X_u \times X_v\|} = \frac{(-f_u, -f_v, 1)}{\sqrt{1 + f_u^2 + f_v^2}}$$

Example.

Consider the upper hemisphere

$$X: D \rightarrow \mathbb{R}^3: (u, v) \mapsto (u, v, \sqrt{1-u^2-v^2})$$

Let a unit speed curve in X

$$\gamma: (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow X[D]: t \mapsto (\sin t, 0, \cos t).$$

The Frenet frame is

$$T(t) = (\cos t, 0, -\sin t)$$

$$K(s) = 1$$

$$f_u = \frac{-2u}{2 \cdot f} = -\frac{1}{f}, \quad f_v = -\frac{1}{f}.$$

$$N(t) = (-\sin t, 0, -\cos t)$$

$$\mathcal{L}(s) = 0.$$

$$B(t) = (0, 1, 0).$$

$$f_u = -u/f, \quad f_v = -v/f$$

And the normal

$$n(t) = (u/f, v/f, 1) / \underbrace{\sqrt{1 + \frac{u^2+v^2}{f^2}}}_{=f}$$

$$1 + \frac{u^2+v^2}{f^2} = \frac{1-u^2-v^2}{f^2}$$

$$= (u, v, f).$$

The normal curvature at $\gamma(t) = (\sin t, 0, \cos t)$ is

$$K_n = \langle T', n \rangle = \langle (-\sin t, 0, -\cos t), (\sin t, 0, \cos t) \rangle = -1.$$

But, since the curvature $K^2 = K_n^2 + K_g^2$, so $K_g = 0$.